Team Details

- **Team name** – Aero-Controllers
- **Team leader name** – Sanjay Kumar S
- **Problem Statement** - Autonomous drones operating in GPS denied environments suffer cumulative INS drift from IMU bias, noise, and integration errors, necessitating a lightweight, real time learning based system capable of stabilizing flight under disturbances, suppressing drift, and maintaining bounded rotational and translational stability without external correction.

Brief about the idea

- This project proposes a **Learning-Aided INS Drift Suppression Framework for GPS-Denied Quadrotor Flight**.

The solution consists of two stages:

- **Phase 1–3 (Completed):**
  A Proximal Policy Optimization (PPO)-based reinforcement learning stabilizer trained up to 600k steps to ensure robust mid-air stability under deterministic and stochastic wind disturbances.

- **Phase 4 (Upcoming):**
  Integration of a lightweight neural bias estimator into a Sensor Fusion pipeline (EKF) to predict IMU bias corrections (Δbg, Δba) and suppress inertial drift in real time.

The system aims to achieve:

- 30% reduction in drift error compared to classical INS+EKF
- Stable mid-flight behavior across multiple disturbance regimes
- Real-time deployment capability on embedded systems

Opportunities

This solution enables reliable drone operation in **GPS-denied environments**, unlocking applications in:

- Indoor inspection (warehouses, factories, power plants)
- Disaster response in collapsed structures
- Underground mining and tunnels
- Military and defense surveillance
- Urban canyon navigation
- Smart infrastructure monitoring

By suppressing INS drift and ensuring stable mid-air control, the system improves reliability, safety, and autonomy in environments where GPS is unavailable or unreliable.

How Will It Solve the Problem?

Reinforcement Learning Stabilization

- A PPO-trained policy (600k steps) learns optimal control actions to counter wind disturbances and maintain bounded roll, pitch, and angular velocity in real time.

Learning-Aided Drift Suppression (Upcoming)

- A lightweight neural bias estimator predicts IMU bias corrections (Δbg, Δba), which are fused using an Extended Kalman Filter (EKF) to reduce cumulative inertial drift.

Result

- Stable mid-air behavior + reduced long-horizon drift → Enables reliable autonomous flight without GPS.

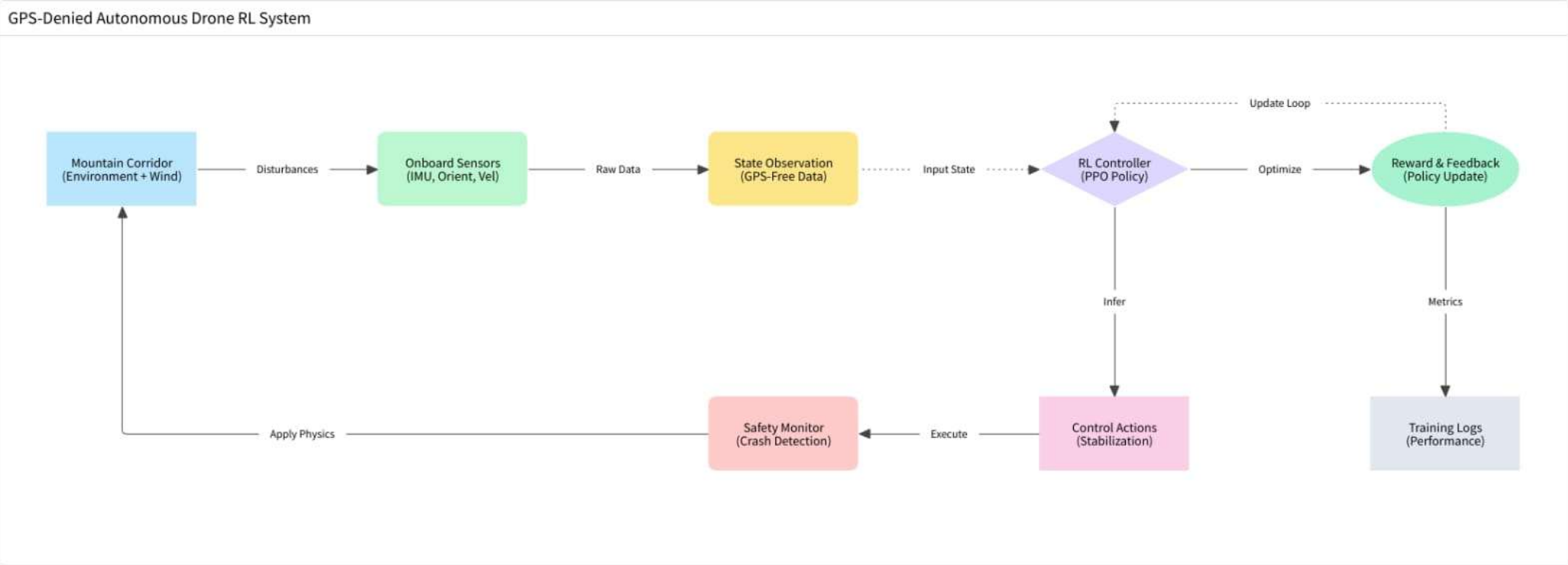
How Different Is It? (USP)

- Unlike traditional INS+EKF systems that assume static sensor bias models, this solution:
- Uses reinforcement learning for robust disturbance stabilization
- Learns dynamic IMU bias corrections from flight data
- Adapts to stochastic wind conditions
- Maintains bounded rotational stability under strong disturbances
- Is validated across calm, mixed, and strong regimes
- Is tested in a physics-accurate Unreal Engine 5 + Cosys AirSim environment

List of features offered by the solution

- PPO-based robust mid-air stabilization
- Multi-regime wind disturbance handling (calm, mixed, strong)
- 600k-step trained reinforcement learning policy
- Deterministic evaluation mode for consistent performance
- Real-time sensor logging (IMU, barometer, angular velocity)
- Stability scoring and quantitative performance metrics
- Checkpoint-based altitude evaluation
- Physics-accurate simulation in Unreal Engine 5 + Cosys AirSim
- Hybrid CPU–GPU training pipeline
- Upcoming learning-aided INS drift suppression with EKF integration

Process flow diagram



Wireframe

Dashboard Overview

System Status
Active

GPS Status
Unavailable

Environment Type
Mountain

Key Alerts
No alerts

Live Drone Monitoring

Drone Orientation
Roll: 5° | Pitch: 3° | Yaw: 10°

Velocity Indicators
23 m/s

Stability Status
Stable

Wind Disturbance
Low

Training & Learning View

Reinforcement Learning Status
In Progress

Reward Trend

Training Progress

Policy Improvement
Incremental

Sensor & Disturbance View

IMU Data Panels
Accelerometer: OK | Gyroscope: OK

Noise/Disturbance Level
Moderate

Sensor Health Status
All Functional

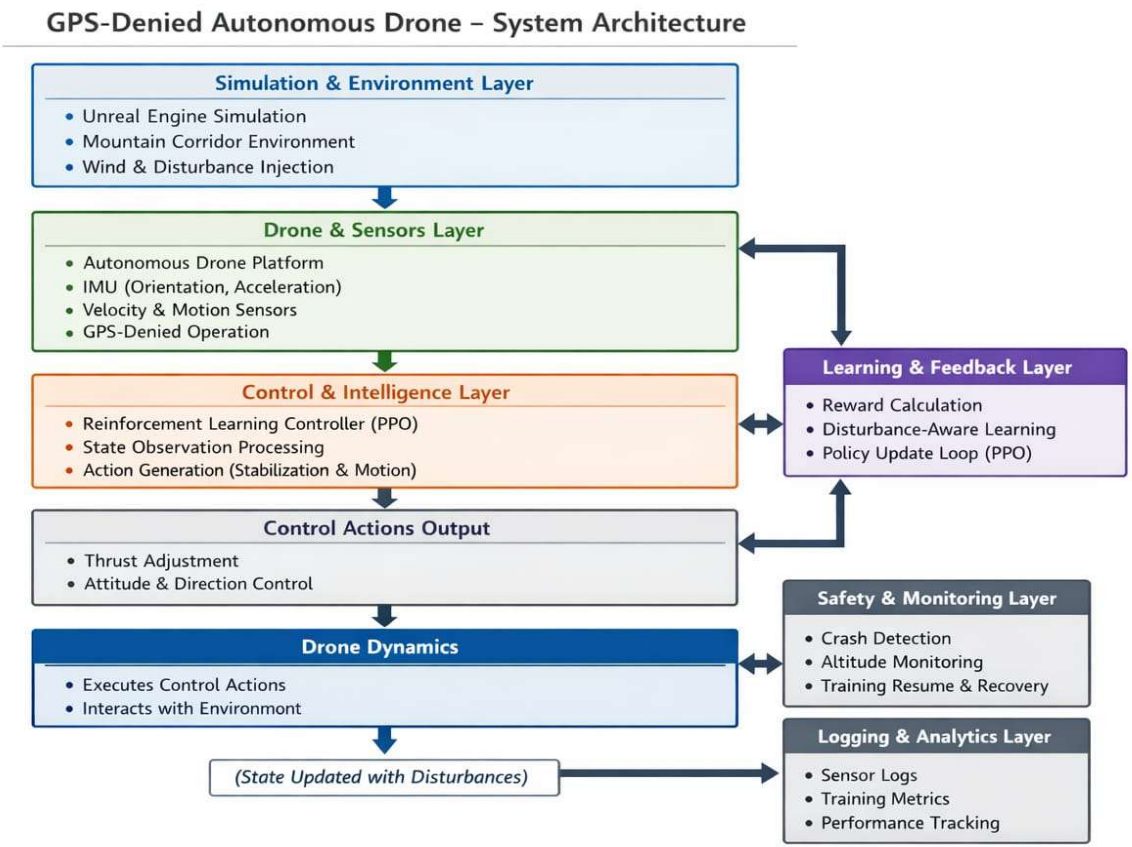
Logs & Safety

Crash Detection Alerts
No recent crashes

Event Logs
Last log: 2023-10-15 14:35:02

Recovery Status
Operational

Architecture diagram of the proposed solution



Technologies to be used in the solution

Simulation Engine

- Unreal Engine 5.5.4
- Cosys-AirSim Plugin (April 2025 release)
- Blocks Editor Test Environment (UE5 Project)

Reinforcement Learning Stack

- Python 3.10
- PyTorch (CUDA-enabled)
- Stable-Baselines3 (PPO Implementation)
- NumPy
- Matplotlib

Build & Toolchain Environment
Windows Build Environment:

- Windows 11
- Visual Studio 2022 Community (v17.13.6)
- Microsoft Visual C++ 14.43.348083
- Windows SDK 10.0.22621.0

Communication & Integration

- RPC Client API (AirSim-Python Interface)
- Real-time sensor streaming (IMU, Barometer, Angular Velocity)
- Closed-loop control pipeline

Usage of AMD Products

- The system utilizes an **AMD Ryzen 5 7600X processor** to handle compute intensive simulation and reinforcement learning workloads.

Key contributions of the AMD platform include:

- High single-core IPC performance for real-time physics stepping
- Multi-threaded processing for environment rollouts
- Efficient handling of Unreal Engine 5 + AirSim simulation loops
- Stable execution of 600k-step PPO training cycles
- The AM5 platform with DDR5 memory provided sufficient bandwidth and memory capacity to support large rollout buffers and policy update cycles.
- This hybrid CPU-driven simulation architecture enabled scalable reinforcement learning training in a high-fidelity drone environment.

Estimated implementation cost

Component (Prototype level)	Estimated Cost
Software Stack (UE5, Python, AirSim)	Open-source
Development Environment	Existing System
Compute Hardware	Already Available
Total (Prototype Level Cost)	Minimal / No Additional Cost

Estimated implementation cost

Component (Deployment level)	Estimated Cost
Flight Controller (Pixhawk-class)	₹17,000 – ₹21,000
High-Quality IMU Module	₹9,000 – ₹25,000
Companion Computer (Jetson / Mini PC)	₹33,000 – ₹60,000
Frame + Motors + ESC	₹41,000 – ₹75,000
Battery & Power Systems	₹16,000 – ₹33,000
Sensors & Assembly	₹16,000 – ₹33,000
Total (**Deployment Level Cost)**	~₹1.25L – ₹2.5L

Prototype Assets

Github Repository Link : https://github.com/Sanjay1712KSK/Aero-Controllers

Demo Video Link: https://www.youtube.com/watch?v=I__WB5NJ0XI

Interactive Documentation Link: https://sanjay1712ksk.github.io/Aero-Controllers/

Prototype Result Summary (Phase-3 Stability Performance (600k Model))

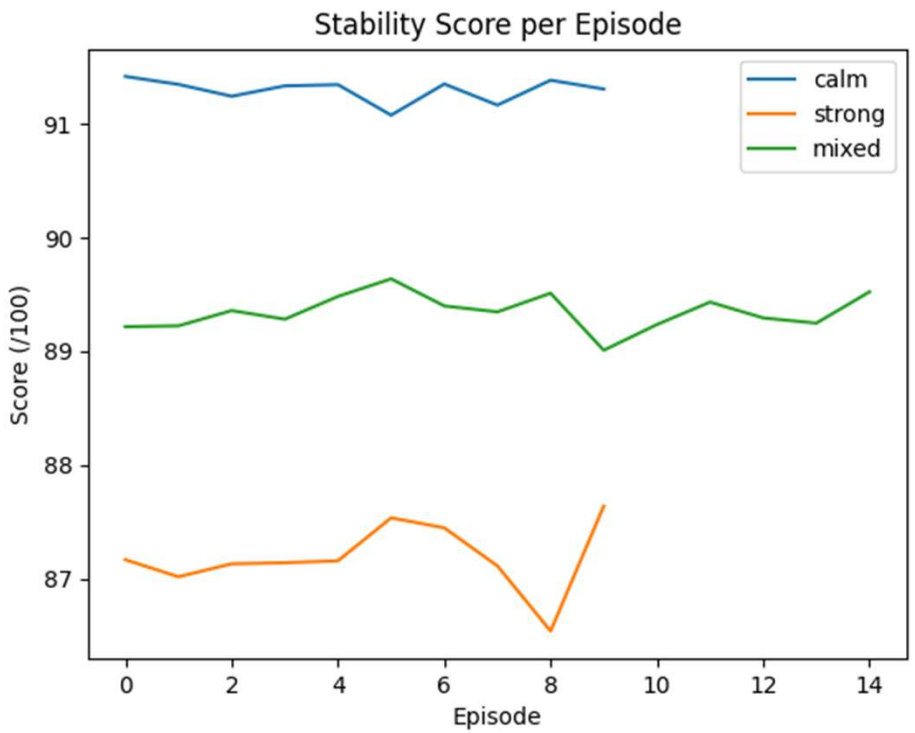
- **Stability Score Across Regimes**
  Calm: 91–92
  Mixed: 89–90
  Strong: 86–87
  Maintained >85/100 across all wind conditions.

- **Bounded Angular Velocity**
  Rotational magnitude remained within stable limits (no divergence or runaway oscillations).

- **Stable Attitude Envelope**
  Roll & pitch maintained within ±0.2 rad under stochastic disturbances.

- **Multi-Regime Robustness**
  Consistent performance across calm, mixed, and strong wind distributions with no degradation trend.

- **Deterministic Convergence Behaviour**
  Evaluation mode showed repeatable, stable control with no instability growth across 200-step episodes.



AMD Slingshot

HUMAN *IMAGINATION*
BUILT WITH *AI*

Powered by H2S HACK2SKILL

Thank you!